SHANA YUNSHENG LI

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EDUCATION

University of Illinois Urbana-Champaign 2024 - Now
Ph.D. in Mathematics Advisor: Nathan Dunfield
Southern University of Science and Technology 2022 - 2024
M.Sc in Mathematics Advisor: Stavros Garoufalidis
Southern University of Science and Technology 2018 - 2022
B.Sc in Mathematics and Applied Mathematics with Honors

RESEARCH INTEREST

Low-dimensional Topology; Mathematical Physics; Quantum Topology; Knot Theory; Hyperbolic 3-Manifolds.

ACADEMIC VISITS

Simons Laufer Mathematical Sciences Institute Jan. 2026 - Feb. 2026
Program Associate Topological and Geometric Structures in Low Dimensions

GRANTS & FUNDING

ACCESS Discover Allocation (MTH260046, 1,500,000 ACCESS Credits) 2026 - 2027
Principal Investigator The 12-tetrahedra census of orientable cusped hyperbolic 3-manifolds

PUBLICATIONS

9. (with Stavros Garoufalidis) The colored Jones polynomial does not determine the Links–Gould polynomial, 10 pages, arXiv:2610.07198

8. (with Marc Kegel and Qiuyu Ren) Small undecidable groups and unrecognizable 4-manifolds, 14 pages, arXiv:2609.10461.

7. (with Stavros Garoufalidis) The FK_3 knot polynomial, 14 pages, arXiv:209.01506.

6. (with Nathan Dunfield, Marc Kegel and Qiuyu Ren) Trisection invariants of 4-manifolds are uncomputable, 6 pages, arXiv:2608.27811.

5. (with Stavros Garoufalidis and Josephine Yu) Positivity and concavity of the colored Links–Gould polynomials of knots, 17 pages, arXiv:2605.30754.

4. Fixed-parameter tractable computation of Reshetikhin–Turaev knot polynomials via tensor networks, 13 pages, arXiv:2606.29662.

3. The complete 10-tetrahedra census of orientable cusped hyperbolic 3-manifolds, Proceedings of the 42nd International Symposium on Computational Geometry (SoCG 2026), Volume 367, pages 73:1-73:15. Full version: arXiv:2512.02142.

2. (with Stavros Garoufalidis) Patterns of the V_2-polynomial of knots, Experimental Mathematics (2026), 1-31. arXiv:2409.03557.

1. (with Ni An and Stavros Garoufalidis) Algebraic aspects of holomorphic quantum modular forms, Research in Mathematical Sciences. 11 (2024) no. 3, Paper No. 54, 1-20.

PROJECTS

4. (with Stavros Garoufalidis) The loop expansion of colored knot polynomials for Lie superalgebras. Ongoing
3. Extending tabulation of quasi-alternating links. Ongoing
2. Census of 12 tetrahedra oriented cusped hyperbolic 3-manifolds. Ongoing
1. Census of 11 tetrahedra oriented cusped hyperbolic 3-manifolds. Drafting

THESES

M.Sc On the Quantum Modularity Conjecture for Knots
B.Sc Foundation of Supergeometry and Its Application in Mathematical Physics

TALKS

Oct. 2026 Invited talk at AMS 2026 Fall Central Sectional Meeting at University of Minnesota-Duluth, Special Session on Computation in Low-dimensional Topology.
Topic: Fixed-parameter tractable computation of Reshetikhin–Turaev knot polynomials via tensor networks.

Sep. 2026 Seminar talk at Group, Geometry and Topology Seminar (UIUC).
Topic: Computations in low-dimensional topology: quantum knot polynomials and ribbon concordance.

May. 2026 Invited online talk at Graduate Topology Seminar (SUSTech).
Topic: Introduction to Reshetikhin–Turaev knot polynomials.

Mar. 2026 Seminar talk at Graduate Geometry-Topology Seminar (UIUC).
Topic: The complete census of orientable cusped hyperbolic 3-manifolds, up to 11 tetrahedra.

Feb. 2026 Invited seminar talk at Computation in Geometric Topology, Topological and Geometric Structures in Low Dimensions, at Simons Laufer Mathematical Sciences Institute (SLMath).
Topic: The complete census of orientable cusped hyperbolic 3-manifolds, up to 11 tetrahedra.

Nov. 2025 Invited online talk at Australian Geometric Topology Webinar.
Topic: Multivariable knot polynomials, the V_n-polynomials, and their patterns.

Oct. 2025 Invited talk at AMS 2025 Fall Central Sectional Meeting at Saint Louis University, Special Session on Foliations and 3-manifolds.
Topic: The complete census of orientable cusped hyperbolic 3-manifolds, up to 10 tetrahedra.

Sep. 2025 Online talk at Equivariant Topological Quantum Field Theory workshop at Casa Mathemática Oaxaca (CMO).
Topic: Multivariable knot polynomials, the V_n-polynomials, and their patterns.

Sep. 2025 Seminar talk at Group, Geometry and Topology Seminar (UIUC).
Topic: Multivariable knot polynomials, the V_n-polynomials, and their patterns.

May. 2025 Conference talk at GTA Philadelphia 2025, 10th annual Graduate Student Conference in Algebra, Geometry, and Topology (GSCAGT) at Temple University
Topic: Multivariable knot polynomials, the V_n-polynomials, and their patterns.

Apr. 2025 Invited online talk at Shanghai Institute for Mathematics and Interdisciplinary Sciences (SIMIS).
Topic: Multivariable knot polynomials, the V_n-polynomials, and their patterns.

Mar. 2025 Seminar talk at Graduate Geometry-Topology Seminar (UIUC).
Topic: Multivariable knot polynomials, the V_n-polynomials, and their patterns.

Mar. 2024	Seminar talk at Graduate Topology Seminar (SUSTech). Topic: The Teichmüller TQFT and the quantum modularity conjecture.

TEACHING

Fall 2026	Teaching assistant of MATH 241 (Calculus III)
Spring 2025	IML graduate leader of undergraduate research project Visualizing Spaces of Curves in 3-manifolds.
Autumn 2022	Teaching assistant of MA109 (Advanced Linear Algebra).

AWARDS

Spring 2024	Outstanding Graduate Graduates Award Outstanding Master Thesis Award
Autumn 2022	Outstanding Teaching Assistant Award
Spring 2022	Outstanding Undergraduate Graduates Award Outstanding Undergraduate Thesis Award

OUTREACH - PIANO PERFORMANCES

Autumn 2023	"Moment · Eternity" Concert	Steinway Garden in Shenzhen Concert Hall
Spring 2022	"Youth Poetry and Music" Chamber Music Concert	Shenzhen Concert Hall
Autumn 2021	National Science & Technology University Aesthetic Education Seminar Concert	Steinway Garden in Shenzhen Concert Hall
Autumn 2020	Lang Lang's Piano Masterclass	SUSTech 10th Anniversary
Autumn 2019	Haochen Zhang's Piano Masterclass	